Final Project Documentation

Project Title: FinTrust Digital Bank-Customer and Transaction Analytics

Project Overview

This project was completed as part of the AnalystLab Africa Experience Lab. It focused on analysing customer behaviour, transaction performance, digital channel usage and risk review patterns using a synthetic digital banking dataset.

The project combined SQL, Python and Power BI to transform raw data into business insights that could support decision-making.

Project Objectives

- Understand customer distribution and engagement across segments.
- Analyse transaction volumes, values and transaction types.
- Evaluate the performance of digital and traditional transaction channels.
- Examine monthly transaction trends and success rates.
- Identify patterns in transactions flagged for risk review.
- Develop an interactive dashboard for monitoring key performance indicators.
- Validate major findings and provide practical business recommendations.

Tools and Technologies

- SQL: Data querying, aggregation, joins, customer segmentation and transaction analysis.
- Python: Data cleaning, exploratory data analysis, statistical summaries and visualisation using Pandas, NumPy, Matplotlib and Seaborn.
- Power BI: Interactive dashboards, KPI measures, visual analysis and slicers.
- Jupyter Notebook: Documentation and execution of Python analysis.
- GitHub: Project version control and documentation.

Project Workflow

1. Data preparation: Loaded the customer and transaction datasets and reviewed their structure, data types and quality.

2. SQL analysis: Queried customer segments, transaction types, channel performance, monthly activity, high-value transactions and risk review patterns.

3. Exploratory data analysis: Used Python to explore distributions, customer behaviour, transaction trends and relationships in the data.

4. Business insights: Identified patterns in customer activity, channel success rates, transaction values and risk review rates.

5. Validation: Compared selected findings against supporting counts, calculations and analytical outputs.

6. Dashboard development: Refined the Power BI dashboard to present customer KPIs, transaction KPIs, trends, segments, channels, transaction status and risk-related patterns.

7. Documentation: Recorded the data quality checks, KPI definitions, business findings, limitations and project workflow.

Key Findings

- Everyday customers formed the largest segment, with 711 customers.
- Transfers recorded the highest transaction volume, with 3,549 transactions.
- The Mobile App was the most frequently used channel, with 5,102 transactions.
- The overall transaction success rate was approximately 90.47%.
- International transactions had a higher risk review rate than domestic transactions in the analysed dataset.
- SMEs had the highest average transaction value per active customer among the identified segments.
- Transaction activity declined in February before recovering in March.

These findings describe patterns in the synthetic dataset and should not be treated as proof of fraud or as evidence of actual banking performance.

Business Recommendations

- Investigate Mobile App transaction failures to identify possible service improvements.
- Review international transaction patterns to understand the factors associated with higher risk review rates.
- Monitor transfers and cash withdrawals, which showed relatively high risk review rates.
- Explore ways to improve services for SME customers based on their transaction behaviour.
- Continue monitoring monthly transaction activity and success rates.
- Maintain regular data quality checks and validate dashboard KPIs before reporting results.

Limitations

- The analysis used synthetic rather than real banking data.
- Missing device and location values limited some comparisons.
- Potential outliers were identified statistically but were not automatically treated as errors.
- Risk review flags did not establish whether transactions were fraudulent.
- Findings were based on the available dataset and should be validated against additional data before real-world decisions are made.

Conclusion

The project demonstrated how SQL, Python and Power BI could be used together to analyse customer and transaction data. The work produced a set of business findings, validation checks and an interactive dashboard to support performance monitoring and further investigation.